

Vatsal Vaghasiya

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MTech Data Science student with two first-author research papers under review and 13 shipped projects spanning LLMs, computer vision, and MLOps. Builds ML systems end to end: data pipelines, training, evaluation, deployment, and drift monitoring. Seeking AI/ML engineering internship or entry-level role.

TECHNICAL SKILLS

Languages: Python, SQL, Dart, Swift
ML / Deep Learning: PyTorch, scikit-learn, XGBoost, KerasNLP, sentence-transformers, OpenCV, MediaPipe
LLMs & RAG: retrieval pipelines, embeddings, vector search (ChromaDB), prompt engineering, Hugging Face Inference API, Whisper
MLOps & Backend: FastAPI, Docker, Docker Compose, Streamlit, drift detection (PSI), Git, SQLite
Data: pandas, NumPy, feature engineering, matplotlib, seaborn, Jupyter

RESEARCH (UNDER REVIEW)

Efficient LLM Preference Prediction via Siamese DeBERTa First author

- Built a 71M-parameter Siamese DeBERTa model that predicts human preference between chatbot responses at 98% of state-of-the-art performance while being 127× smaller; trained in 8.4 hours on free-tier T4 GPUs at \$0 cost.
- Eliminated position bias with swap augmentation and corrected overconfidence with temperature scaling ($T=1.20$), improving log loss by 6.2% and accuracy by 3.5 points over baseline.

ProbCLIP-A: Uncertainty-Aware Cross-Modal Retrieval via Probabilistic Adapters First author

- Designed a 4.2M-parameter probabilistic adapter over frozen CLIP (151M) that outputs calibrated uncertainty; achieved 68.9% R@1 with the best calibration among all baselines (ECE 0.062) on Flickr30K.
- Monte Carlo inference flags 36.6% of retrieval failures at an 8.1% false-positive rate, enabling practical human-in-the-loop review for high-stakes retrieval.

PROJECTS

Constitution of India — RAG Question Answering sentence-transformers · ChromaDB · Mistral-7B

- Built retrieval-augmented QA over the Indian Constitution with exact source citations and similarity scores; wrote the retrieval and prompting logic from scratch (~60 lines, no framework).
- Stress-tested with 20 adversarial queries, published the failure analysis, and improved chunking to reach 78% answer accuracy.

IPL Match Outcome Predictor — MLOps Pipeline XGBoost · FastAPI · Docker Compose · Streamlit

- Deployed a three-service prediction system (REST API, drift monitor, health dashboard); the monitor computes Population Stability Index every 5 minutes and flags the model for retraining on drift.
- Engineered 9 features (rolling win rates, venue history, toss decisions) for an XGBoost classifier serving win probabilities.

Cachy — Short-Form Media Knowledge Engine Python · FastAPI · Flutter · faster-whisper

- Built a cross-platform app that converts Reels/Shorts and articles into structured knowledge cards using local Whisper transcription, Tesseract OCR, and an LLM chain with automatic fallback across three providers; cards linked in a semantic knowledge graph, fully offline-capable.

Bangalore Air Quality Hotspot Analysis pandas · scikit-learn

- Consolidated a year of non-standardized data from 14 CPCB monitoring stations, engineered 6 domain features, and compared K-Means, hierarchical clustering, and DBSCAN; DBSCAN correctly isolated critical hotspots (Silk Board, max AQI 500).

Selected additional projects: AirSwipe (gesture-controlled presentations, MediaPipe/OpenCV), ScribbleType (on-device handwriting recognition, ML Kit), and 7 more on GitHub.

EDUCATION

MTech, Data Science — Ramaiah University of Applied Sciences, Bengaluru Nov 2025 – 2027 (expected)

BTech, Computer Engineering — SAL College of Engineering Oct 2021 – Apr 2025